

Hop-Area Is an Energy Proxy, Not a Second First-Law Term

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Abstract

Jacobson and FGHMV need two pieces: $\delta S = \eta \delta A + \beta \delta E$. A conformal hop bump changes both. This note splits the variations.

Matter only, hops fixed: $\delta A \equiv 0$, $\rho(\delta S, P_{\text{CHM}}) = 0.9998$. The first law does not need area.

Geometry only: $\rho(\delta A, P) = 0.996$. Area and energy are the same bump. A raw partial correlation of -0.987 is the degeneracy artifact of that collinearity and is not scored as independence.

Auditor, $N = 10$: $\rho(A, P) = 0.988$, independence REFUTED.

Papers 39–41 saw δS track area because they saw it track energy. Not $8\pi G$. Not Einstein. Not de Sitter.

Admission. *I asked whether area was a second term. It is not. I did not promote a two-column residual drop under $\rho = 0.996$ into Jacobson.*